Chi-Nguyen Tran

Trần Chí Nguyên

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Education

B.S. in Artificial Intelligence September 2023-October 2027 (expected)
VNUHCM, University of Science (HCMUS)
GPA: 3.86/4.00 (9.20/10); 134 accumulated credits

Research Interests

I am currently doing research in robotics. My goal is to help robots understand what is around them, know where they are, and work out how to act, and whether it is safe to act, when their surroundings change. So far I have worked on robots that locate themselves in bad weather, agents that act on distorted camera input, and testing for self-driving cars. At PhD level, I would like to go deeper into teams of robots and sensors: when one member quietly sees or performs worse than it appears, adding it can make the whole team worse, and I want to find how a team could learn from evidence how much to trust each member.

Research Experience

AI R&D Lead, Realtime Robotics February 2026-Present
Leading research on perception and localization models for robots, including localization without satellite signals on drones, and on running these models on the small embedded computers that robots carry, under tight compute and sensor constraints.

Selected Research Outputs

An asterisk marks equal contribution as indicated in each cited work; my name is shown in bold; paper titles are hyperlinked to their arXiv or DOI record.

NeurIPS 2026 (accepted)

Weather-Robust Cross-View Geo-Localization via Prototype-Based Semantic Part Discovery. *NeurIPS 2026, Main Track, poster.*

Chi-Nguyen Tran*, Duy-Minh Dao-Sy*, Trung-Kiet Huynh*, Phu-Hoa Pham, Phu-Quy Nguyen-Lam, and Long Tran-Thanh.

- Developed a lightweight prototype-based semantic-part head for patch-based Vision Transformers, achieving state-of-the-art weather-robust drone-to-satellite geo-localization with the fewest parameters among top-performing methods on SUES-200, University-1652, and DenseUAV under ten weather-corruption conditions. Co-first author, 1st of 3.

MIST: Reliable Streaming Decision Trees for Online Class-Incremental Learning via McDiarmid Bound. *NeurIPS 2026, Main Track, poster.*

Phu-Hoa Pham*, **Chi-Nguyen Tran***, Phu-Quy Nguyen-Lam, Duy-Minh Dao-Sy, Trung-Kiet Huynh, and Long Tran-Thanh.

- Streaming decision trees that learn new classes online from a data stream, with split decisions backed by a McDiarmid concentration bound. Co-first author, 2nd of 2.

Co-author on three further NeurIPS 2026 papers:

- *Training-Free Cultural Alignment of Large Language Models via Persona Disagreement.* Main Track, poster; 4th of 8 authors.
- *Unlocking Compositional Generalization in Continual Few-Shot Learning.* Main Track, poster; 4th of 6 authors.
- *Visual Robustness and Neural Alignment in a Shared Foraging Task: The Mouse vs. AI Benchmark.* Evaluations and Datasets Track, poster; 8th of 14 authors, written with the organizers as the first-place team.

Journal (accepted)

Payoff Scaling Shapes Cooperation in LLM Agents across Languages. *Interface Focus (The Royal Society), Q1 journal, accepted November 2026.*

Trung-Kiet Huynh, Duy-Minh Dao-Sy, …, **Chi-Nguyen Tran**, …, and The Anh Han (16 authors).

- How the size of the stake and the prompt language change the strategies LLM agents play in cooperation dilemmas. Co-author, 11th of 16.

Peer-reviewed publications

RoadFury at the ICST 2026 Tool Competition: Self-Driving Car Testing Track. *IEEE ICST 2026, pp. 509-510, DOI 10.1109/ICST69053.2026.00071.*

Chi-Nguyen Tran*, Duy-Minh Dao-Sy*, Trung-Kiet Huynh*, Phu-Hoa Pham, and Phu-Quy Nguyen-Lam.

- Developed a regression-test prioritization method for self-driving-car simulation testing that surfaces failure-revealing test cases earlier. Co-first author, 1st of 3.

MEMRES: A Memory-Augmented Resolver with Confidence Cascade for Agentic Python Dependency Resolution. *FSE Companion 2026, pp. 1438-1441, DOI 10.1145/3803437.3808242.*

Chi-Nguyen Tran*, Duy-Minh Dao-Sy*, Trung-Kiet Huynh*, Phu-Hoa Pham, Phu-Quy Nguyen-Lam, and Vu Nguyen.

- Developed a memory-augmented confidence cascade for Python dependency resolution that escalates unresolved conflicts to language models selectively. Co-first author, 1st of 3.

Selected Honors

Competition wins

- **1st Place** (both tracks), NeurIPS 2025 Mouse vs. AI Robust Foraging Competition (international), team HCMUS_TheFangs. 2025
- **Winner**, FSE-AIWare 2026 Agentic Python Dependency Resolution Competition, team MEMRES. 2026
- **Third Prize**, Vietnam Student Artificial Intelligence Olympiad National Final; **Second Prize**, Southern Regional Round. 2025

Other recognitions

- **Top 3.3%** (138/4,182 teams), NVIDIA Nemotron Model Reasoning Challenge, Kaggle, team HCMUS-TheFangs. 2026

Technical Skills

Programming: Python, C++
Machine Learning: PyTorch; Hugging Face / timm; Vision Transformers; reinforcement learning
Systems and Tools: ONNX, TensorRT, CUDA, distributed training, Linux, Git
Languages: Vietnamese (native); English (VSTEP 7.0/10, CEFR B2)

Professional Service

Reviewer, ACL and CVPR workshops (2026).